Image Forgery Detection Using ResNet50 Classification + U-Net Segmentation with XAI (Grad-CAM) Analysis

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Abstract

The rapid advancement of digital image editing tools and AI-based content generation has significantly increased the prevalence of image manipulation, raising serious concerns in journalism, digital forensics, law enforcement, and online media verification. This paper presents an explainable deep learning framework for image forgery detection and localization that integrates classification, segmentation, and visual interpretability into a unified pipeline. The proposed system employs ResNet50 for binary classification of images into real and forged categories. For images detected as forged, U-Net is used to localize manipulated regions at the pixel level. To enhance forensic transparency, Grad-CAM is incorporated to visualize decision-relevant regions. The framework is evaluated on the CASIA 2.0, achieving reliable classification performance with high forgery recall and meaningful localization accuracy despite the small size of manipulated regions. The results highlight the importance of combining detection, localization, and explainability for trustworthy forensic image analysis.

I. Introduction

Digital image manipulation has become increasingly accessible due to the widespread availability of powerful editing software and AI-based generative models, leading to the growing misuse of forged images for misinformation, social manipulation, and fabrication of legal evidence. As visual content continues to play a critical role in societal decision-making, the need for reliable automatic image forgery detection has become a major research challenge in computer vision and cybersecurity. Traditional hand-crafted feature-based methods often struggle to generalize across diverse and sophisticated manipulation techniques. In contrast, deep learning-based approaches have demonstrated strong performance by learning hierarchical visual representations directly from data [5]. Architectures such as deep residual networks have achieved state-of-the-art performance in image classification tasks and are well-suited for forgery detection [1]. However, most deep learning models operate as black-box systems, which limits their trustworthiness in forensic applications. Recent explainable AI techniques, such as Grad-CAM, enable visual interpretation of deep model decisions and help address this critical limitation [3]. Motivated by these challenges, this work proposes an end-to-end explainable deep learning framework that integrates image-level forgery detection, pixel-level localization, and visual interpretability into a single unified system for trustworthy forensic analysis.

II. Dataset

This study utilizes the CASIA 2.0 Image Tampering Dataset [4], which contains over 12,000 authentic and tampered images created using copy-move and splicing techniques, along with binary ground-truth masks for a subset of forged images. For classification, images are labeled as real (0) or forged (1) and divided using stratified sampling into 70% training, 15% validation, and 15% testing to preserve class distribution. The training set includes 5,243 real images and 4,002 forged images, indicating a moderate class imbalance, which is handled using a class-weighted loss function. For segmentation, only forged images with available masks are used, and a filename normalization strategy is applied to correctly pair images with their corresponding masks. These image-mask pairs are split into 80% training and 20% validation. Due to noisy annotations and the very small size of manipulated regions, pixel-level localization remains a challenging task.

III. Methods

The proposed framework consists of three sequential components: image-level classification, pixel-level segmentation, and visual explainability. For classification, a ResNet50-based convolutional neural network pretrained on ImageNet is used as the backbone [1]. The original fully connected layer is removed and replaced with a custom binary classification head consisting of a fully connected layer (2048-256), batch normalization, ReLU activation, dropout, and a final output layer (256-1). The classifier is trained using BCEWithLogits loss with class-weighting (pos_weight) to address class imbalance between real and forged images. Data augmentation techniques including rotation, cropping, color jitter, and blur are applied to improve robustness and generalization. For images classified as forged, pixel-level localization is performed using a U-Net segmentation model trained with a combined binary cross-entropy and Dice loss to handle severe foreground-background imbalance [2]. For visual explainability, Grad-CAM is applied to the trained classifier to generate heatmaps that highlight image regions most influential to the model's predictions, improving forensic transparency and trust [3].

IV. Experimental Setup

All experiments are conducted using standardized image resizing and normalization. The classifier is trained for multiple epochs using the Adam optimizer with learning-rate scheduling, while the segmentation network is trained independently using paired image-mask samples. Model selection is based on validation performance. Classification performance is evaluated using accuracy, AUC, recall, and F1-score, while segmentation performance is evaluated using Dice and IoU metrics. Grad-CAM visualizations are generated only for the trained classifier during inference to analyze decision behavior.

V. Results and Discussion

The proposed framework is evaluated on the CASIA 2.0 dataset [4] using standard classification and segmentation metrics. The ResNet50-based classifier [1] achieves an overall accuracy of 80.32% with an AUC of 0.86, demonstrating strong discriminative capability between real and forged images. A forged-image recall of 0.88 indicates that the model successfully detects most manipulated images, which is critical for forensic applications where missed forgeries are highly undesirable. The F1-score of 0.80 confirms a balanced trade-off between precision and recall. For pixel-level localization, the U-Net-based segmentation model [2] achieves a Dice score of approximately 0.21 and an IoU score of approximately 0.15. Although these values appear low, they are consistent with prior observations on

CASIA due to the extremely small size of manipulated regions and the presence of noisy ground-truth masks. Visual explainability provided by Grad-CAM [3] reveals that the classifier primarily focuses on texture inconsistencies, boundary artifacts, and splicing regions, confirming that predictions are based on meaningful forensic cues rather than random correlations.

A. Classifier Training Dynamics

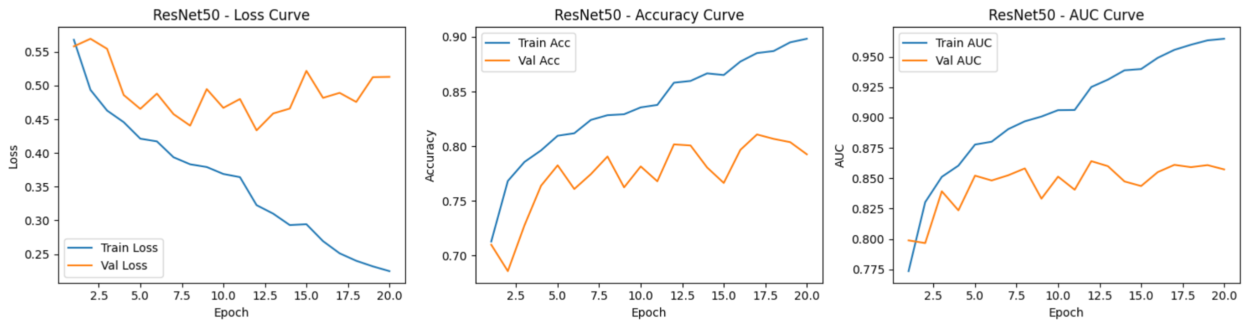

Figure 1. ResNet50 training curves: loss, accuracy, and AUC over 20 epochs (train vs. validation).

Training loss falls steadily (0.56 to 0.22) and training accuracy climbs to about 0.90 over 20 epochs. Validation loss drops early and then plateaus around 0.44-0.52 with fluctuations after epoch 8, while validation accuracy settles near 0.79-0.81 and validation AUC holds at approximately 0.86. The widening train/validation gap indicates mild overfitting: the model keeps memorizing training samples after validation stops improving. The final checkpoint is therefore selected on validation performance rather than taken from the last epoch.

B. Test Confusion Matrix

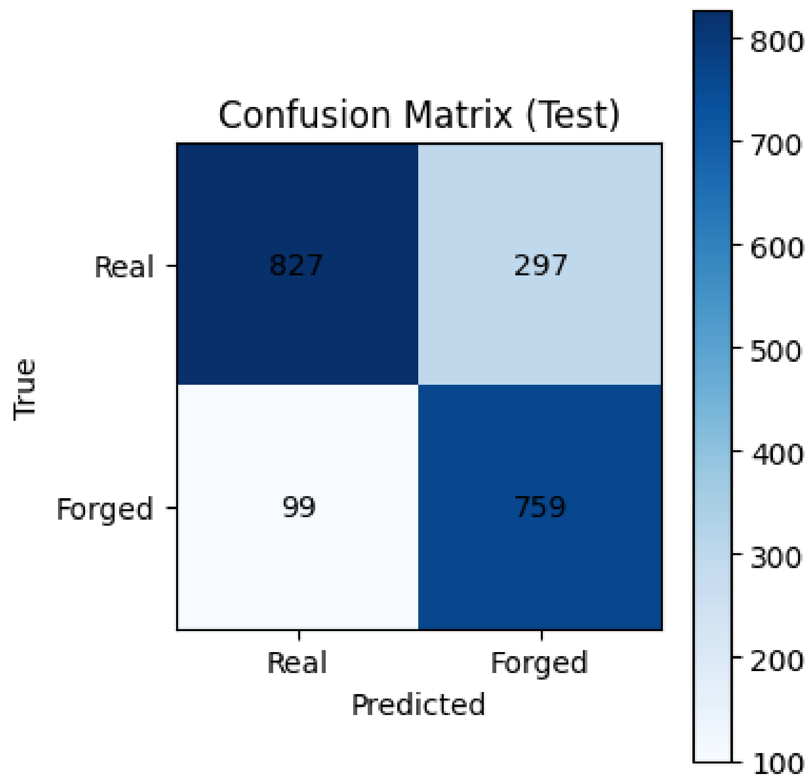

Figure 2. Confusion matrix on the 1,982-image test set (1,124 real / 858 forged).

The test confusion matrix records 827 true negatives, 297 false positives, 99 false negatives, and 759 true positives, giving an accuracy of 80.0% (1,586/1,982), consistent with the reported 80.32%. Forged-image recall is 88.5% (759/858): only 99 forgeries are missed. Forged-image precision is 71.9% (759/1,056), for an F1 of about 0.79. False positives outnumber false negatives roughly 3-to-1. For a forensic screening tool this trade-off is correct: a flagged real image receives human review, while a missed forgery does not.

C. Segmentation Training Dynamics

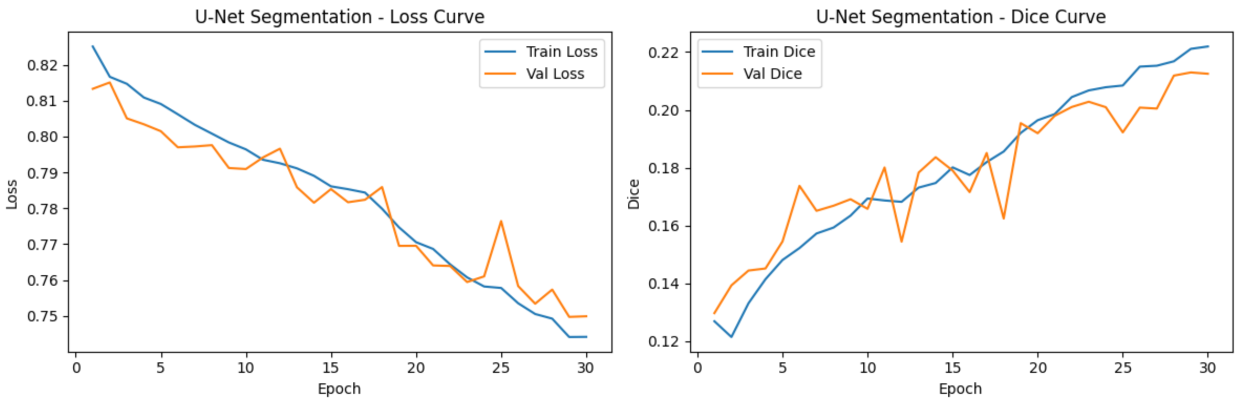

Figure 3. U-Net training curves: loss and Dice score over 30 epochs (train vs. validation).

Over 30 epochs, training and validation loss decline together (0.82 to about 0.75) and the Dice score climbs from roughly 0.13 to 0.22 (train) and 0.21 (validation). Validation Dice is noisy, as expected with tiny forged regions and imperfect masks, but it tracks the training curve upward without diverging, indicating genuine learning rather than overfitting. The low absolute Dice reflects a data ceiling, not a training failure: when a tampered region spans only a few dozen pixels and the ground-truth masks are noisy, even a good localizer scores low.

D. Explainability Spot-Checks

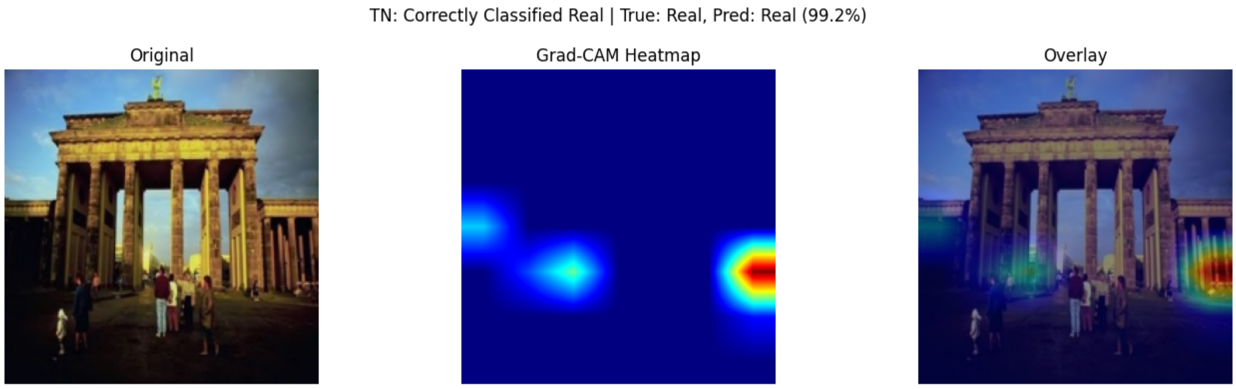

Figure 4. Grad-CAM for a correctly classified real image (99.2% confidence): sparse, localized activation with no coherent tampering pattern.

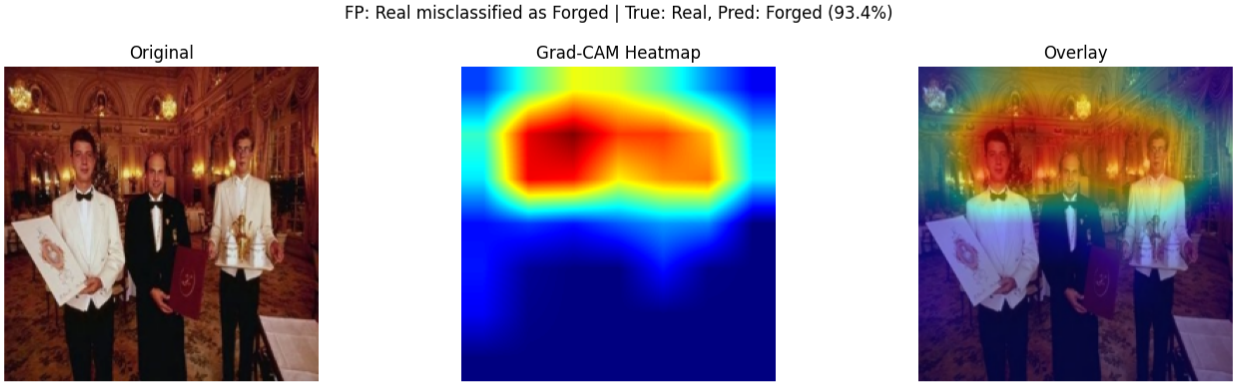

Figure 5. Grad-CAM for a real image misclassified as forged (93.4%): intense activation over faces and the ornate interior.

The true-negative example shows sparse, localized activation with no coherent tampering pattern: nothing in the image looks spliced, and the heatmap agrees. The false-positive example is more instructive. The heatmap fires intensely across faces and the ornate interior, where strong lighting contrasts, chandeliers, and fine facial textures mimic the boundary and texture artifacts the model associates with tampering. This is precisely the failure mode discussed in Section VI: sensitivity to complex natural textures and lighting. Grad-CAM makes that failure auditable instead of silent, which is the point of including explainability in a forensic pipeline.

VI. Limitations

Although the proposed framework demonstrates reliable classification and meaningful localization performance, several limitations remain. First, pixel-level forgery localization using U-Net is strongly affected by the noisy and incomplete ground-truth masks provided in the CASIA 2.0, as well as by the extremely small relative size of manipulated regions, which significantly constrains the achievable Dice and IoU scores. Second, the segmentation network is trained from scratch without a pretrained encoder, which limits its ability to learn robust spatial features compared to transfer-learning-based approaches. Third, the ResNet50 classifier occasionally produces false positives in regions with strong shadows, lighting variations, and complex textures, indicating sensitivity to certain natural image artifacts. Finally, the dataset primarily contains traditional copy-move and splicing forgeries, and therefore the model's generalization to deepfakes and AI-generated images remains limited, motivating the need for broader training data in future work.

VII. Conclusion and Future Work

This paper presents a complete explainable deep learning framework for image forgery detection, localization, and interpretation. The system demonstrates reliable classification performance with high forgery recall and meaningful visual explanations, making it suitable for forensic verification tasks. A key limitation of the current system is that localization accuracy is constrained by noisy ground-truth masks and the extremely small size of forged regions. Future work will focus on improving segmentation performance using pretrained encoders, extending the framework to deepfake and AI-generated image datasets, and deploying the system as a real-time forensic verification tool.

References

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